Engineering Software with Agentic AI

Published companion reconciliation
Completion report | Edition 2026.10.23-d11

The book now incorporates the published companion and its reported MacBook validation. All eleven actions in this revision are complete. The accepted book, current learning guide and refreshed extracts are bound to the final content-acceptance record; the previously accepted figure sources are preserved.

Accepted output	Recorded result
Main book	419 physical PDF pages; 23 native figure occurrences
Advanced Learning Protocol	30 pages; 2 figure occurrences; current edition source and two identical delivery copies
Sample and appendix extracts	Sample: 52 pages. Appendix: 12 pages. Current table of contents also refreshed.
Figures and changed layouts	24 distinct TikZ bodies; 25 captioned and text-referenced occurrences; 4 layout findings resolved
External reader evidence	R17 / CR10 remains OPEN_EXTERNAL

What changed for the reader

The opening instructions now start at the public repository root and create a separate study copy. Prerequisites distinguish the observed Mac and Linux routes. Chapter 4 explains the guided Mac CSV repair without merging it with the older implementation and verification episodes. The book also points to the current learning guide and retained historical evidence.

Reviewed public commit

`e58e0f3eeec5b4531d65d90fac7cb9948edad731`

Published repository at this commit | Getting started | Execution audit

The repository was inspected without remote mutation. This report makes no claim of a GitHub commit, push, publisher submission or production deployment.

Technical evidence: three different scopes

Observation	Environment and result	What it establishes
Author-reported Mac audit	macOS 26.5.1 arm64; Python 3.12.4; Git 2.56.0. 518 core tests; overall PASS, with 1 explicit Linux real-clock skip. 11 extended tests passed.	A dated execution report for the specified Mac route. It does not show that the real Linux owner-clock check executed on macOS.
Author-reported Linux core rerun	Linux ARM64 in Docker 27.3.1; Python 3.12.15; Git 2.47.3. All 518 core tests passed with 0 skips; real owner clock executed. 11 extended tests passed.	A separate ARM64 core run. The optional emulated Linux x86-64 model runtime is another environment.
This revision's bounded entry verification	21 commands; 14 chapter cards; Linux, Python 3.12.14, git version 2.51.1. Doctor clean before/after; new Chapter 1 copy and existing Chapter 2 destination refusal checked.	Expected faulty baseline: 7 cases, 3 passes, 2 failures, 2 errors, exit 1. Entry/source integrity, not a repeat of the author's full Mac audit.

Published audit: scopes and results | Audit limits and reproduction

Read the counts correctly

The report's categories overlap, and the 17-check CSV reviewer invokes the seven-case contract checker. They are not 24 unique requirements. The initial native-regression table records 15 tests; a later follow-up records 17 after additional launcher and RSS-unit fixes. Earlier stages remain historical.

Native follow-up and 17-test result

Practical setup now stated in the book

The core declares Python 3.10 or later; observed routes use Python 3.12. Use Git with merge-tree --write-tree support. On macOS, use canonical study paths, TMPDIR=/private/tmp and Python -B (or PYTHONDONTWRITEBYTECODE=1). Run doctor from the directory containing reader_start.py and require empty protected-file and release-starter findings before preparing a new sibling copy.

The bounded entry receipt records 3,443 tracked files with unchanged bytes and executable bits matching the pinned Git objects. It made no model calls and did not execute the full 518-test suite.

Local evidence: revision_d11/evidence/entry/ENTRY_VERIFICATION.json; COMMANDS.json; source_review/BOOK_CONTENT_REVIEW.json.

The native CSV evidence has three episodes

These observations share a selected local CSV contract. They have different candidates or stopping conditions, explicit assistance and separate budgets. Keep those distinctions when interpreting the result.

1. Original native CSV implementation

4 CLI invocations; four-call allowance exhausted. 2,700 client seconds allowed; 1,274.40 used.

OPEN_PARTIAL_BUDGET_EXHAUSTED. The third invocation passed seven native cases but overstated cleanup. The fourth cleaned the file without a fresh native check. Independent final review passed 17 selected checks, including the seven-case checker.

Assistance and limit: Author corrective guidance; supervised teaching observation. Independent checks do not supply a missing native tool event; preserve this episode's original status.

Pinned primary record

2. Earlier separate verification-only continuation

1 additional CLI invocation; fifth across the first two episodes. Separate allowance of at most 3 calls/1,800 client seconds; one call used 202.64792135199968 seconds.

Actual unchanged checker ran on the final candidate; seven cases passed, the handoff accurately identified final bytes and scope, and the client stopped. All eight candidate files stayed unchanged; separate 17-check review passed.

Assistance and limit: Author supplied the exact verification command; no candidate edits permitted. The public URL documents the historical gap, not the later completion; the completion evidence is edition-local. One CLI invocation made three backend requests.

Public historical context

Completion evidence: publication_evidence/native_csv_completion_20261023/INDEX.md and review/NATIVE_CONTINUATION_ACCEPTANCE.json.

3. Later fresh guided MacBook CSV repair

2 successful-route invocations: implementation then documents-only continuation; preceding failed attempts remain separately recorded. Each invocation retained its 900-second deadline; reported elapsed times 24.856 and 27.330 seconds.

First invocation passed behavior but only 15/17 independent checks because spec/plan were missing. Documents-only continuation kept the API hash unchanged. Final 7/7 selected contract and 17/17 independent checks, with both demos exiting 0.

Assistance and limit: Explicit guidance to read API/selection/contract, preserve signature/JSON, and use csv.writer with io.StringIO; effective read configuration corrected; no supplied reference patch/manual candidate edit. Guided technical pass; the plan still has a stale finding. Seven checks overlap with the seventeen. Not an autonomous success, perfect prose, human acceptance or portable publication of the raw packet.

Pinned primary record

The later Mac run used OpenCode 1.19.3, Qwen3-4B Q4_K_M, native Darwin ARM64 llama.cpp b11799 and Metal on the observed M2 Max/64 GiB workstation. Its published workflow does not promise identical future output or timing.

Guided workflow and machine scope | Guidance, document review and outside-Git packet

Completed actions | R01-R06

The following dispositions are read directly from the final current ACTION_REGISTER.json. They record this book update; earlier revision actions retain their own histories.

R01 | COMPLETE | Publish the companion locator and commit-bound evidence references

Public companion URL, pinned checkout command and three dated primary citations added.

Evidence: manuscript/frontmatter/reading_note.tex | Pinned primary source

R02 | COMPLETE | Update prerequisites and distinguish observed macOS and Linux test scopes

Core Python floor retained; published macOS and separate Linux ARM64 reports distinguished; host-specific real clock not relabeled portable.

Evidence: manuscript/frontmatter/reading_note.tex | Pinned primary source

R03 | COMPLETE | Make Chapter 1 and Chapter 2 entry commands work from the repository companion root

Chapters 1 and 2 use reader_start.py from the companion root with unused sibling working directories. All 14 show commands, doctor and intended Chapter 1 baseline independently checked.

Evidence: manuscript/chapters/chapter01.tex | Pinned primary source

R04 | COMPLETE | Document native Apple Silicon setup and explicit runtime selection

Apple Silicon setup guide and explicit runtime selection added, separating native Metal from emulated execution.

Evidence: manuscript/chapters/chapter02.tex | Pinned primary source

R05 | COMPLETE | Add the separately guided Mac CSV outcome and distinguish behavioral and document review

Separate guided Mac CSV outcome added with failures, explicit API/read-scope guidance, documentation-only continuation and overlapping-check limits.

Evidence: manuscript/additions/ch04_native_complete.tex | Pinned primary source

R06 | COMPLETE | Preserve and correctly locate the earlier native completion evidence omitted from the public checkout

All 86 missing historical continuation files retained byte-for-byte, with wrapper map to three original pinned public context links.

Evidence: publication_evidence/README.txt | Pinned primary source

Completed actions | R07-R11

R07 | COMPLETE | Supply the current canonical advanced-learning guide and preserve its Linux real-clock boundary

Current canonical guide rebuilt and visually accepted at 30 pages; identical copies supplied beside the book and in publication_evidence. Participant exposure and Linux real-clock boundaries retained.

Evidence: manuscript/supplements/advanced_learning.tex | Pinned primary source

R08 | COMPLETE | Align bibliography, source guide, title, edition metadata and current source package entry

Edition, bibliography, source guide, book-package entry and companion-relative paths aligned; all five location-review findings resolved.

Evidence: README.txt | Pinned primary source

R09 | COMPLETE | Verify changed entry commands and source-to-manuscript claims

Independent content review passed; bounded entry-command evidence passed; location defects resolved without altering recorded historical packets.

Evidence: revision_d11/source_review/BOOK_CONTENT_REVIEW.json | Pinned primary source

R10 | COMPLETE | Rebuild main book and supplement; preserve all native figure captions and prose references

Main book (419 pages) and supplement (30 pages) rebuilt from frozen sources; all 25 captions and numbered prose references verified. All 24 TikZ bodies and 25 figure environments are unchanged from d10; final visual checks pass.

Evidence: revision_d11/builds/build03/BUILD_RESULT.json

R11 | COMPLETE | Refresh extracts, review changed layouts and verify final publication package

The final 419-page book yields an exact 52-page Chapters 1/7 sample, 12-page Appendix B and current TOC text. Every extracted page and retained link is verified; all four new layout findings are resolved. Publication contents are accepted; report and ordinary-ZIP integrity are recorded separately in the final report QA and adjacent delivery receipt.

Evidence: revision_d11/FINAL_CONTENT_ACCEPTANCE.json

Current guide and retained evidence

The accepted 30-page d11 guide uses the canonical manuscript and refactored TikZ sources. The reviewed public checkout still contains the older 29-page PDF; its editable-source README identifies edition 2026.10.21-d4. The current guide is supplied beside the book and in publication_evidence/.

The earlier verification-only continuation preserves 86 historical files (307,019 bytes). Its wrapper maps original context links to the pinned public repository. It is separate from the new Mac candidate and raw packet, which remain outside Git.

Older public guide source identity

Figure preservation and resolved layouts

Main book	Learning guide	Distinct TikZ bodies	Captions and callouts
23	2	24	25

Every canonical TikZ body matches accepted d10 bytes. The shared transfer drawing accounts for the difference between 24 distinct bodies and 25 uses. The integrated audit binds each caption, stable label and numbered prose reference to the accepted main-book or supplement PDF and its frozen sources.

Resolved defects found in changed-page review

Finding	Observed issue	Verified correction
LAY-01	Chapter 1 first setup listing was split after its first line; doctor was isolated at the bottom of page 32.	Reserve fourteen baselines before Step 1 to keep its heading, introduction and five-line setup block together. Disposition: RESOLVED
LAY-02	The projection-list introduction ended page 105 while all three items began page 106.	Reserve six baselines before the list introduction so the first item follows it. Disposition: RESOLVED
LAY-03	Listing 7.9 caption wrapped its final word alone onto a second line.	Shorten caption to Run the harness from a fresh companion checkout; cumulative-work preservation remains in adjacent prose. Disposition: RESOLVED
LAY-04	Chapter 1 optional Codex paragraph began with one line at the foot of page 39; its instruction continued on page 40.	Reserve four baselines before the run-in heading to keep its opening instruction together. Disposition: RESOLVED

Acceptance coverage

The final content record contains 8 accepted checks. Source inventory, source-to-report review, bounded entry evidence, figure integration and the recorded visual/extract checks remain separately inspectable. The final integrated audit records no blocking build diagnostics.

The retained build records still identify inherited KOMA/typearea/listings warnings where applicable. Their presence is not reported as missing figure captions. Changed-page visual review is not a fresh exhaustive review of every inherited page or every historical runtime example.

Evidence: revision_d11/FINAL_CONTENT_ACCEPTANCE.json; INTEGRATED_FIGURE_AUDIT.json; FIGURE_SOURCE_AUDIT.json; CURRENT_SOURCE_RECONCILIATION.json; LAYOUT_FINDINGS.json; visual_qa/.

Optional repository maintenance and external work

The five follow-ups below concern a future repository maintenance commit. This book update cites the newer specific report and retains edition-only material separately. No remote change is claimed.

UP01 | Refresh the landing page and runtime summary

Add a dated current summary and report link for the 23 October Mac/Linux audit, with 518 core tests, the one Mac Linux-clock skip and separate environments; retain historical rows as history and fill the public repository locator.

Do not convert the observed machines into blanket Windows, Intel Mac or cross-interpreter certification.

Relevant public record

UP02 | Distinguish the 15-test and 17-test native stages

Label the initial 15-test stage as historical and add a current summary of the later 17-test result, linking the follow-up.

Preserve chronology; do not rewrite old receipts or sum 15 and 17.

Relevant public record

UP03 | Publish the current learning guide with its source

In a future repository refresh, publish the current PDF and its canonical LaTeX/TikZ source layout together, with a clear edition label and entry link.

Keep the participant exposure notice and the Linux real-clock boundary; final d11 build/delivery remains a separate gate.

Relevant public record

UP04 | Expose the separate verification-only continuation

Publish or link the preserved continuation as a separate episode and provide navigation from the original journey.

Keep the original four-call disposition and budgets unchanged; do not conflate the continuation with the later Mac repair.

Relevant public record

UP05 | Consider a portable subset of the new Mac evidence

If the author elects to publish a portable evidence subset, retain candidate/prompt/effective-configuration identities, failed stages, session exports, independent checks and precise chronology; link it from the report and guide.

This is an optional future evidence-publication action. The d11 book must continue to attribute the current narrative report without claiming access to unavailable raw packets.

Relevant public record

R17 / CR10 | OPEN_EXTERNAL

Author technical execution does not supply independent intended-reader observations. Independent intended-reader observations, actual learning and delayed transfer require their own participants and review. Technical execution and synthetic rehearsal cannot establish team field economics or production outcomes.

The real learning pilot retains its Linux kernel boot-ID and same-host/boot requirements. A participant must not read the transfer case, worked answer or reference answers before an unseen-transfer assessment. Record prior exposure as guided study rather than treating it as an unseen response.

What the author audit does not establish

Artifact identity and reproducibility

The following SHA-256 identities are read from final acceptance and checked against the actual delivered files before report generation. PDF page counts are also verified. The JSON companion contains the complete action, check, source and build records used by this report.

Main book | 419 pages | build03

`Engineering_Software_with_Agentic_AI_Draft_20261023_d11.pdf`

`SHA-256 35a52377c6b0dd8098cec1c50cb1e0f4d77864bc121e6d1dc2a946945cbaea09`

Advanced Learning Protocol | 30 pages | build01

`Advanced_Learning_Protocol_20261023_d11.pdf`

`SHA-256 3836f076c2456be6fd437d3721c45885a2d4a1d694ebd393c4b5d4ecb7418392`

Sample extract | 52 pages

`Engineering_Software_with_Agentic_AI_Sample_Chapters_01_07_20261023_d11.pdf`

`SHA-256 7eae45e14f6c2e75d1c27ca74b3ecbdc35ceb7c57238710a514935a7b38f43be`

Appendix extract | 12 pages

`Engineering_Software_with_Agentic_AI_Appendix_B_20261023_d11.pdf`

`SHA-256 6bd47954b1f898beb3cbd1746eb2bed6989aff477016c38ebcb51bf1e5e8688f`

Table of contents

`Engineering_Software_with_Agentic_AI_Table_of_Contents_20261023_d11.txt`

`SHA-256 8ad162a4f6a90dd437e9d68b4524a171c7d585bb721eb00544ac56df0fa3a3ce`

Rebuild from the source package

Use XeLaTeX and latexmk with the manuscript's fonts/packages. The captured environment used Biber 2.19 and BibLaTeX 3.19. Temporary dependency paths are environment provenance, not required reader paths.

```
bash manuscript/build.sh
bash manuscript/build_supplement.sh
python3 -B revision_d11/audit_figures.py
python3 -B revision_d11/build_completion_report.py --check-inputs
python3 -B revision_d11/build_completion_report.py
```

The report builder requires final content acceptance PASS and R01-R11 COMPLETE. It verifies accepted artifact hashes and page counts, current figure/build binding and resolved layout records; it does not rerun laboratory suites.

Report and archive identity

The same-stem JSON records this report PDF's SHA-256 and all input identities. It does not contain its own hash. The final archive is verified separately by the adjacent delivery receipt after packaging. No final ZIP verification is asserted by this report or its generated JSON.

Pinned companion | Execution audit | Guided CSV workflow